

InsightOS - Practical Hackathon Presentation & Q&A Cheat Sheet

1. PROJECT OVERVIEW

* **What is InsightOS?** An AI-powered Business Intelligence platform designed for small and medium retail businesses in India (clothing stores, pharmacies, restaurants, grocery shops).

* **Why**
margins

30-Second Elevator Pitch

"Small retailers in India lose lakhs every year to stockouts, expired inventory, and low margins because they can't analyze their raw CSV data. **InsightOS** is a zero-backend BI platform where business owners simply upload their POS CSV files to instantly discover what's wrong, how much money they're losing, and get step-by-step action plans. Powered by **IBM Granite AI**, it provides data-grounded answers to natural-language business questions - running free, private, and 100% inside the browser."

2. KEY FEATURES

1. **Fuzzy CSV Data Ingestion** - Normalizes messy CSV headers from any POS system automatically (e.g., "MRP" or "Sales Amt" ? "revenue"). **Why it matters:** Business owners can upload files directly without manually reformatting columns.

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glance.

3. TECHNOLOGY STACK

Technology	Used For	Why We Chose It	Alternative Considered
React 19	Frontend UI Framework	Declarative component architecture	Vue.js (smaller ecosystem for map/c
TypeScript	Type Safety	Enforces strict data contracts for	Plain JavaScript (high risk of runt
Vite	Build Tool & Dev Server	Sub-second HMR and built-in API	Webpack (slower build times and com
TailwindCSS	UI Styling	Rapid utility-first styling for dar	Vanilla CSS (slower development ite

Recharts	Data Visualizations	Declarative React charting library	D3.js (overkill and requires low-le
React-Leaflet	Store Heatmap	Open-source map library requiring	Google Maps (requires credit card &
PapaParse	CSV Parsing	Handles client-side CSV parsing, m	SheetJS (heavier library than neede
IBM Granite	AI Copilot (LLM)	Enterprise-focused LLM; runs local	OpenAI GPT-4 (expensive API fees an

4. IBM TECHNOLOGY

* ****Which IBM Technology?**** IBM Granite LLM (`granite3-dense:8b` locally via Ollama / `ibm/granite-13b-chat-v2` via IBM watsonx.ai cloud API).

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5. ARCHITECTURE

Actual System Flow:

****User ? React 19 Frontend ? PapaParse / CSV Normalizer ? Deterministic Rules Engine (dataAnalysis.ts) ? Recharts / Leaflet UI & IBM Granite AI Copilot****

* ****Frontend (React 19 + Vite):**** Single Page Application rendering the 5 dashboard views.

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6. AI PIPELINE

Important Flow:

****CSV File Upload ? Header Normalization ? Deterministic Rules Analysis ? Context-Injected Prompt ? IBM Granite LLM ? Natural Language Insights****

* ****How AI Contributes:**** The deterministic rules engine finds **what** happened (e.g., "Bengaluru store margin is 12%"). The IBM Granite AI Copilot explains **why** and advises **what to do** in plain English based on the full business context.

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7. COMPETITORS

Competitor	InsightOS Advantage
Excel / Google Sheets	InsightOS automatically surfaces hi
Power BI / Tableau	InsightOS is zero-cost, zero-setup,
Existing POS Systems	POS systems only record raw transac

Generic AI (ChatGPT)	ChatGPT lacks business data and vis
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8. TOP JUDGE QUESTIONS

1. **Q: What problem are you solving?**

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9. HARD QUESTIONS

1. **[!] Why not just use Excel?**

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problem

10. LIMITATIONS & FUTURE ROADMAP

3 Current Limitations:

1. **No Data Persistence:** Data lives in React state and resets to sample data when the browser is refreshed.

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3 Future Improvements:

1. **Database & Auth Integration:** Add PostgreSQL + OAuth2 (IBM App ID) for persistent multi-store accounts.

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11. 8-SLIDE PRESENTATION STRUCTURE

* **Slide 1: Title & Elevator Pitch**

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12. QUICK REVISION BEFORE PRESENTATION

* **Project in 1 sentence:** InsightOS is a zero-backend BI platform that turns raw retail CSV data into actionable business alerts and IBM Granite AI insights.

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